

FIFA World Cup 2026 Tournament Network

Introduction

The 2026 FIFA World Cup is jointly hosted by the United States, Canada, and Mexico. During the tournament, six participating national football teams establish dedicated communication headquarters to coordinate player management, match scheduling, medical services, scouting, logistics, media operations, and official communications. For security and administrative purposes, each team's headquarters operates as an independent network while remaining connected through the FIFA Tournament Backbone Network. In this project, you will design and implement the complete network infrastructure connecting the following six team headquarters:

- Argentina
- Brazil
- England
- France
- Germany
- Spain

The estimated number of network users in each headquarters is shown below. These values must be used when preparing the VLSM addressing plan.

Team Headquarters	Estimated Users
Brazil	280
Argentina	210
France	170
England	130
Spain	90
Germany	60

1. Topology Design

Build the topology by following all of the conditions below.

- Each team headquarters must be represented by a separate router.
- The Spain headquarters must be connected only to the France headquarters.
- The Brazil, Argentina, and France routers must be connected through one common Layer-2 switch.

- A direct connection must be established between the France and England routers.
- A direct connection must also be established between the France and Germany routers.
- England and Germany must be directly connected to each other to provide an alternative communication path.
- Each headquarters must contain its own LAN switch.
- Only two PCs may be used to represent all users of each headquarters.
- Each headquarters must contain one network printer.

2. Server and Service Configuration

DNS Server

The FIFA tournament will use one centralized DNS server.

- The DNS server must be located in the France headquarters.
- Every headquarters must use this DNS server.

Web Servers

Dedicated web servers must be installed in the following headquarters.

- Brazil www.brazil2026.com
- Germany www.germany2026.com

Every PC in the tournament network must be able to access both websites using their domain names.

Email Servers

Separate email servers must be installed in

- Argentina
- England

The email domains are

- argentina2026.com
- england2026.com

Each email server must contain at least two user accounts.

Example

- captain@argentina2026.com
- coach@argentina2026.com
- captain@england2026.com
- medic@england2026.com

Users from the Argentina domain must be able to exchange email with users from the England domain in both directions.

3. DHCP Configuration

The tournament network uses both router-based and server-based DHCP.

- The France router must provide DHCP services for
 - France

- Brazil
- Argentina
- Spain
- England must use a dedicated DHCP server.
- Germany must use a dedicated DHCP server.

All PCs must receive their addresses dynamically. Routers, servers, printers, and switches must use manually configured IP addresses.

4. Addressing and VLSM

Base Network Address

The base network address must be generated from the first student's ID.

Take the last four digits of the student ID.

Split them into two groups of two digits.

Use

- First pair → First octet
- Second pair → Second octet

The third and fourth octets must be zero.

Use a /16 network.

5. Demonstration

Each group member must explain their assigned work.

The demonstration must include

- Successful VLSM implementation
- DHCP verification
- Website access through DNS
- Cross-domain email communication
- Static routing verification
- Dynamic routing verification
- Floating static route failover by disabling the primary link

Student ID :22131547

Last four digits : 1547

Base network : 15.47.0.0/16

If the first pair is 00, replace it with 10.

Students must perform complete VLSM subnetting for

- All LANs
- All point-to-point WAN links

5. Routing

France Router: The France router must perform hybrid routing.

It must

- Participate in RIPv2 with Brazil and Argentina.
- Use static routes to reach
 - Spain
 - England
 - Germany

Spain Router

Spain communicates with the tournament through France only.

Configure

- One default static route pointing toward France.

England Router

Configure recursive (next-hop) static routes for all remote headquarters through France.

Germany Router

Configure directly attached static routes for all remote headquarters through France.

Only the outgoing interface may be specified.

Do not configure fully specified static routes.

6. Floating Static Routes and Backup Paths

The direct connection between England and Germany acts as a backup path.

England Backup

Under normal operation, England reaches the rest of the tournament network through France.

If the France–England link fails,

- England must forward traffic through Germany.
- This backup route must appear only after the primary route becomes unavailable.

Germany Backup

Under normal operation,

Germany reaches the tournament network through France.

If the France–Germany link fails,

- Germany must forward traffic through England.
- The backup route must not replace the primary route while the primary connection remains active.

Each failure scenario must be tested separately.

7. Connectivity Requirements

The completed network must satisfy the following conditions.

- Every PC must obtain its IP address automatically.
- Every headquarters must be able to communicate with every other headquarters.

- Every PC must access both websites using DNS names.
- Cross-domain email communication must function correctly.
- Network printers must be reachable from PCs within the same headquarters.
- DNS name resolution must function throughout the network.
- Primary and backup routing paths must operate correctly during link failures.

You are allowed to make any valid and necessary assumptions while designing the network infrastructure.

Deliverables

- The network mentioned above should be implemented in Cisco Packet Tracer, with the necessary devices and full configuration.
- **In your topology, you must show the network address of each network by placing notes in Cisco Packet Tracer.**
- After completion, you should be able to test the conditions imposed, and all the devices should be able to ping one another.
- You will have to submit the following:
 - Work Distribution among the group members [Who did which part]
 - **As a member of the group, you must configure at least one of these 4 setups: DHCP/Email/Static Routing/Dynamic Routing.**
Then, you can divide the rest of the work among yourselves as you see fit.
 - The pkt/pka file
 - Picture of the Network topology diagram with proper labels [You have to show the network addresses using notes for each network]
 - A PDF containing
 - VLSM tree
 - IP address table
 - The configuration commands of all the routers you have implemented.